

Neuropsychological Functioning Associated with High-Altitude Exposure

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This article focuses on neuropsychological functioning at moderate, high, and extreme altitude. This article summarizes the available literature on respiratory, circulatory, and brain determinants on adaptation to hypoxia that are hypothesized to be responsible for neuropsychological impairment due to altitude. Effects on sleep are also described. At central level, periventricular focal damages (leuko-araiosis) and cortical atrophy have been observed. Frontal lobe and middle temporal lobe alterations are also presumed. A review is provided regarding the effects on psychomotor performance, perception, learning, memory, language, cognitive flexibility, and metamemory. Increase of reaction time and latency of P300 are observed. Reduced thresholds of tact, smell, pain, and taste, together with somesthetic illusions and visual hallucinations have been reported. Impairment in codification and short-term memory are especially noticeable above 6,000 m. Alterations in accuracy and motor speed are identified at lower altitudes. Deficits in verbal fluency, language production, cognitive fluency, and metamemory are also detected. The moderating effects of personality variables over the above-mentioned processes are discussed. Finally, methodological flaws found in the literature are detailed and some applied proposals are suggested.

KEY WORDS: altitude; acute hypoxia; hypocapnia; brain dysfunction; mild neuropsychological impairment.

This article reviews current developments in research on neuropsychological functioning in altitude with a particular focus on biological and environmental factors. Attention is paid to hypoxia as a responsible stimulus for the consequences of altitude together with other modulating factors such as individual differences. Additionally, methodological shortcomings in this kind of research are also discussed.

In 403 BC the Chinese traveler Fa Hsein reported the first documentary record of the effects of altitude on

human beings, in this case probably lung edema (Gilbert, 1983). In 326 BC Plutarch described several symptoms of mountain sickness in his chronicles of the march of Alexander over India (Plutarch, 1912). Fifteen centuries later, in 1298, Rustinian noted down several malaises suffered by Marco Polo during his explorations of Tibet (Castelló-Roca, 1993). Jose de Acosta, a Spanish Jesuit of the sixteenth century spoke in his book *Historia natural y moral de las Indias (Natural and Moral History of the West Indies)* about his sickness, vomits, and headaches while on a visit to the Andean mountain range (De Acosta, 1590). Acosta's seems to have been the first known case of acute mountain sickness (AMS) although there is no complete agreement on whether it was AMS or gastroenteritis. Yet, the first clinical classification of this disorder was made by Thomas Ravenhill, a British surgeon who worked in several mines in the north of Chile (Ravenhill, 1913). The AMS is an acute syndrome associated with an inadequate adaptation to a hypoxic environment. AMS is a syndrome with unspecific symptoms, most characteristically headache, insomnia, dizziness, fatigue, and

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Table 1. The Lake Louise Consensus Committee Definition of Acute Mountain Sickness

Diagnostic feature	Description
Criteria	A recent gain in altitude At least several hours at the new altitude Presence of headache
Symptoms	Gastrointestinal upset (i.e., lack of appetite, nausea or vomiting) Fatigue or weakness Dizziness or lightheadedness Difficulty sleeping

Note. In order to diagnose AMS, all criteria 1, 2, and 3 and one of symptoms are required. Adapted from "The Lake Louise Acute Mountain Sickness Scoring System," by Roach, R., Bärtsch, P., and Oelz, O. (1993). In: Sutton, J., Houston, G., and Coates, G. (eds.), *Hypoxia and Molecular Biology* (pp. 272–274), Queen City Printers, Burlington, VT.

gastrointestinal symptoms (lack of appetite, nausea, or vomiting). AMS is observed in up to 25% of persons who have recently reached an altitude of 2,000 m or more (Montgomery et al., 1989). The most widely accepted set of diagnostic features was established by the Lake Louise Consensus Committee and is presented in Table 1 (Roach et al., 1993).

Apart from western references, AMS has always been identified in populations living at high altitude. Indigenous Quechua call it *soroxtchi* and Tibetans named it *mundara*. AMS drew the attention of members of expeditions to high mountains and has been the object of thorough biomedical research in the last 30 years (Hornbein et al., 1989; West, 1984). As a result, advances have been made in the identification of its physiopathological foundations (Basnyat and Murdoch, 2003; Garrido and Botella, 1998; Hornbein and Schoene, 2001).

Corporatively, neuropsychological impairment has been the focus of fewer studies. Neuropsychological impairment in high altitude were sited very early by Janssen (1890). On an ascent to Mont-Blanc, this author experienced symptoms of AMS when observed that a sudden weakness prevented him from noting down the symptoms of AMS. This was overcome by a quick series of aspirations. Accordingly, before thinking he made several aspirations. Janssen concluded that intellectual capacity required strength and oxygen so that to preserve it, the climb had to be made without making any physical effort (Jansen, 1890). No other precedent is found until 1911, when Massot i Palmers gave a paper at the I Catalanian Conference of Excursionists entitled "El excursionismo como medio del desarrollo físico e intelectual en el niño" (Physical and intellectual development of children by means of excursions; Massot i Palmers, (1911)). There have been numerous anecdotal references

to emotional, cognitive, and perceptive alterations associated with altitude in well-known expeditions (Dunlap, 1918; Herzog, 1952; Ruttledge, 1934; Shipton, 1943; Wilmer and Berens, 1918). Later on, the administration of neuropsychological tests was intended to obtain indirect evidence of brain damage by researchers who were not psychologists and consequently no particular attention was devoted to the functional pattern of alterations. A ground-breaking exception in this respect was the work by McFarland (McFarland, 1932, 1937a, 1937b, 1937c, 1937d, 1941; McFarland and Barach, 1937; McFarland and Evans, 1939). McFarland investigated the altitude-related alterations of visual adaptation to darkness, psychomotor coordination, immediate memory, and the influence of anxiety on oxygen tension. McFarland employed diverse methods to observe the effects of hypoxia: extended periods at high altitudes, fast ascents in aircraft, and hypobaric chambers. Between the 1960s and 1970s, several interesting attempts were made at establishing patterns of neuropsychological impairment associated with altitude (Denison et al., 1966; McFarland, 1971; Ryn, 1971). In the last two decades, more thorough and rigorous studies have related behavior alterations with respiratory and neurological variables (Garrido et al., 1995; Hornbein et al., 1989; Regard et al., 1989; West, 1984).

The present article is specifically centered on the altitude neuropsychological impairment (ANI) in high mountaineering although related areas will also be mentioned including animal models, and chronic hypoxia disorders such as the chronic obstructive pulmonary disease (COPD) or the obstructive sleep apnea (OSA). Chronic hypoxia conditions should be distinguished from acute hypoxia conditions—the duration of acute hypoxia ranges between some seconds and several weeks while chronic hypoxia is characteristic of chronic respiratory disorders or extended permanence in a hypoxic environment. In this article, we refer almost exclusively to cases of acute hypoxia studied in laboratory conditions or in real ascents.

Since the work of McFarland, 100 of studies concerning the effects of altitude have been published. Their methods have been highly diverse so a comprehensive assessment is most complex. The following review is intended to include all relevant evidence with a particular emphasis on the neuropsychological impairment. This article briefly reviews the respiratory, vascular, pathological, (i.e., AMS) and neurological foundations of the neuropsychological functioning at high and extreme altitude. In addition, the moderating role of sleep patterns associated with altitude are also analyzed. Then, we review in more detail the empirical evidence regarding cognitive malfunctioning due to altitude. In this respect, we discuss the likely dysfunction of: (a) psychomotor performance (i.e., psychomotor

speed, psychomotor accuracy, permanent mild psychomotor dysfunction), (b) perceptive processes (i.e., reaction time (RT) studies, perceptive discrimination studies, color perception studies, hallucinatory experiences reports), (c) memory, learning and attention (e.g., retrieval vs. storage alteration, spatial memory, short-term memory), (d) functions associated with frontal lobe (i.e., verbal fluency, cognitive flexibility, metacognition). Afterwards, we consider the moderating role of anxiety and personality factors in the neuropsychological outcomes of subjects exposed to hypoxic environment due to altitude. Throughout the text, we propose a number of theoretical analyses based on the available evidence. Moreover, we argue about methodological shortcomings and challenges addressed by neuropsychological assessment at high altitude. Finally, we suggest a unified ANI assessment strategy.

RESPIRATORY ADAPTATION AND NEUROPSYCHOLOGICAL FUNCTIONING

Exposure to high altitude is considered a case of anoxic (Adams et al., 1999) in which the metabolic deficit is due to a reduced arterial pressure of oxygen. The effect is similar to that observed in chronic form in COPD or in OSA, although, in these cases the environmental pressure of oxygen is adequate and it is the obstruction of the respiratory tract that is responsible for hypoxia.

The proportion of oxygen in the air remains constant at 20.9% up to the limits of the troposphere (approximately 12,000 m). However, pressure decreases exponentially with altitude. The loss of barometric

pressure (P_B), and consequently of oxygen pressure due to altitude is the determining factor of ANI. A pressure of 101 kPa at sea level is cut down to a half at 5,500 m (53 kPa), and to a third at 8,848 m (33.5 kPa). Individuals who ascend high altitudes are subject to a reduction of oxygen in the air they breathe. This phenomenon is called hypoxia. Environmental hypoxia determines the reduction in alveolar pressure of oxygen (PAO_2) and the reduction in arterial partial pressure of oxygen (PaO_2) once this gas passes through the alveolar–arterial interface by diffusion. Hypoxemia is a reduction in oxygen saturation in arterial blood (% SaO_2) associated with hypoxia. For instance, at an altitude of 4,500 m, PaO_2 is 47 mmHg and the saturation is 80% (a reduction of 20%), while at the summit of Mount Everest PAO_2 is 35 mmHg, PaO_2 is 28 mmHg and saturation ranges between 57% and 70% (Sutton et al., 1988; West, 1983b; West, 1984). Extreme reductions above 50% can take place during vigorous exercise at an environmental pressure equivalent to that at the summit of Mount Everest (Sutton et al., 1988). Table 2 shows an extract of the relationships among altitude, P_B , PAO_2 , PaO_2 , and % SaO_2 .

At a high altitude there is a reduction in P_{50} that is the PaO_2 corresponding to a SaO_2 of 50%. This means, that for the same PaO_2 , hemoglobin carries more oxygen. This phenomenon is a result of respiratory alkalosis due to lung hyperventilation produced by high altitude and tends to displace the hemoglobin dissociation curve progressively to the left. As a result of hyperventilation there is a reduction in arterial pressure of carbon dioxide ($PaCO_2$), known as hypocapnia. Hypocapnia produces an intense vasoconstriction in the brain. The result is always positive

Table 2. Respiratory Variables Associated to Oxygen Uptake at Different Altitudes and Barometric Pressures

Altitude		P_B		PIO_2		PAO_2		PaO_2		$PaCO_2$		% SaO_2
M	Feet	Torr	kPa	Torr	kPa	Torr	kPa	torr	kPa	Torr	kPa	
0	0	760	101.3	149	19.9	100	13.3	95	12.6	40	5.3	97
152	500	747	99.3	147	19.5	99	13.2	92	12.2	40	5.3	97
1,610	5,280	640	85.1	123	16.4	84	11.2	77	10.2	34	4.5	95
2,300	7,544	582	77.4	118	15.7	74	9.8	72	9.6	32	4.3	95
4,500	13,725	433	57.7	81	10.8	62	8.2	55	7.3	30	4.0	87
5,500	18,050	379	50.5	69	9.2	38	5.2	36	4.8	26.4	3.5	82
6,300	20,664	351	46.7	59	7.9	35	4.6	27	3.51	9	1.2	61
8,848	29,028	253	33.6	43	5.7	35	4.6	28	3.7	7.5	1.0	75

Note. P_B = barometric pressure, PIO_2 = pressure of inspired oxygen in the trachea, PAO_2 = alveolar pressure of oxygen, PaO_2 = partial pressure of oxygen in arterial blood, $PaCO_2$ = partial pressure of carbon dioxide in arterial blood, % SaO_2 = percent saturation of hemoglobin with oxygen in the arterial blood. These data have been obtained from different investigations with different register situations. Adapted from Martin, L. (1999). *All you really need to know to interpret arterial blood gases*. Lippincott Williams and Wilkins, Philadelphia; West, J. (1984). Human physiology at extreme high altitudes on Mount Everest. *Science* **323**: 784–788; and West, J. B., Hackett, P. H., Maret, K. H., Milledge, J. S., Peters, R. M., Pizzo, C. J., et al., (1983b). Pulmonary gas exchange on the summit of mount Everest. *J. Appl. Physiol.* **55**: 678–687.

for hypoxemia so that the vasomotor tone favors vasodilatation and consequently increases the cerebral blood flow (CBF).

Human beings rely upon several compensatory mechanisms for the hypoxic stimulus. The individual capacity for adaptation to hypoxia by increasing ventilation is established by the evaluation of the hypoxic ventilatory response (HVR). This response is measured as VE40, which means that the volume of exhaled air from normoxic conditions up to conditions of PaO₂ of 40 mmHg (equivalent to an altitude of 5,500 m at rest). Evaluation is made at rest in supine position maintaining isocapnia, that is, constant levels of arterial carbon dioxide (Schoene et al., 1984). The presence and intensity of this response runs parallel to the activity of the nucleus of the carotid body so a causal role has been hypothesized for this structure (see Lahiri et al., 2001). Agadzhanyan et al. (1972) found that damages in the carotid body prevent adaptation to a hypoxia corresponding to 5,000 m and consequently impair the execution of a sequence of conditioned motor responses. It has been recently established that the beginning of HVR displays a time pattern with a similar intensity to the firing rate of the nucleus of the carotid body, whose activation depends on peripheral chemoreceptors that detect metabolic acidosis and decrease of levels of PaO₂ (Lahiri et al., 2000). Recent research emphasizes that extended exposure to high altitude ends up blocking the activity of this nucleus and the peripheral receptors. This coincides with a reduction of HVR and the process of acclimatization (Moore, 2000), and also with the permanent reduction of HVR in highlanders (Moore, 2000). In these communities, this structure shows less sensitivity to environmental hypoxia and a smaller HVR. Besides adaptability to altitude (Lahiri, 1984), HVR seems to have a strong genetic component as suggested by research with twins (Kawakami et al., 1982) and ethnic differences (Moore, 2000).

On the one hand, an intense HVR is a protecting factor against AMS (Hackett et al., 1982) and a predictor of physical adaptation to altitude (Schoene, 1982). On the other hand, subjects with an intense HVR display greater subsequent neuropsychological impairment than subjects with a reduced response during a simulated ascent in a hypobaric chamber to 8,488 m in a 40-day period (Hornbein et al., 1989). The effect seems generalized because of the significant correlations in the said study, which included almost the totality of tests where impairment was found. In particular, significant correlation was reported ($p < .05$) in the simulated ascent group between HVR and Buschke's selective reminding test (SRT)—sum recall—paired associate of the Wechsler memory scale (WMS)—immediate and delayed recall—verbal WMS—immediate

recall—and Aphasia Screening Test. These results were only replicated in the case of the real-ascent group for the number of aphasic errors in the Aphasia Screening Test. This seems to indicate a greater sensitivity of verbal memory and verbal fluency functions to this effect although this interpretation remains an issue for further study.

Hornbein et al. (1989) hypothesized several likely causes of such apparently paradoxical impairment. First, it was thought that subjects showing a high HVR experience lower levels of oxygenation during exercise or sleep thus causing the impairment. However, a number of studies have demonstrated that %SaO₂ of subjects with a weak HVR during exercise is lower than that of subjects with an intense HVR. As a consequence, subjects with a weak HVR should suffer a higher hypoxia triggered by exercise (Oelz et al., 1986; Schoene et al., 1984). Regarding sleep, further evidence has shown that subjects with a high HVR display similar levels of night hypoxia as apposed to those with a weak HVR (Coote et al., 1993a) in spite of showing more periodic breathing (PB) or Cheyne–Stokes breathing during sleep (i.e., a pattern of breathing with varying depth of respiration and brief periods of apnea). As part of Operation Everest II, Hornbein et al. (1989) point out that subjects with a smaller mean %SaO₂ spent most of their sleep time with a smaller saturation compared to the rest and in turn suffered a greater impairment in delayed recall of verbal material established by the WMS. These subjects also showed a greater impairment in the Finger Tapper Test (FTT). Nevertheless, Hornbein et al. (1989) did not find any connection between HVR and the degree of desaturation during the night.

A third explanation of the paradoxical effect (Hornbein et al., 1989) stresses that in strong physical exercise, a high HVR allows greater muscle oxygenation and justifies the better physical performance of these subjects (Masuyama et al., 1986; West, 1984). In addition, the increase of hypocapnia in periods of physical activity should be considered. The vasoconstricting effect of hypocapnia (Gotoh et al., 1965) competes with the increase of CBF and the vasodilating response produced as adaptations to hypoxia (Adams et al., 1999). According to this approach, we consider that the following sequence takes place: first, a high HVR produces high levels of PaO₂, which results in better physical performance and so an elevated hypocapnia. In turn, this results in brain vasoconstriction with an impairment of circulation and oxygenation (ischemia and hypoxia). It is hypothesized that these two factors are responsible for neuropsychological impairment. It has been already mentioned that the ratio between hypoxemia and hypocapnia is usually favorable to hypoxemia in normal subjects as regards to

brain vasodilatation. Nevertheless, we believe that subjects with a high HVR have a higher hypocapnia but this pattern can be reversed at certain times in areas of the encephalon producing a lesser tissue oxygenation.

Further research is needed to associate HVR, neuropsychological impairment, altitude, and rest/activity, in order to check the value of the latter hypothesis, which is supported by indirect evidence. Firstly, subjects with a high HVR show a greater hypocapnia at rest at altitude, and accordingly more brain vasoconstriction associated with it (West et al., 1983a). However, Hornbein et al. (1989) did not find any correlation between the speed of mean blood flow in the middle brain artery during a simulated stay at altitude and HVR or neuropsychological impairment. It should be stressed that these measures were made at rest and that the differences in hypoxemia and hypocapnia between subjects with a high- and low-HVR manifest especially, in exercise.

Recent evidence in agreement with the previous hypothesis, although not validating it directly, is the evolution of HVR in increasing workloads between highland sojourners and adapted subjects. It has been observed that, despite repeated climbs and extended exposure to the highest summits, Sherpas living at altitude show minimal risks of brain damage (Garrido et al., 1996) and the associated memory impairment (Bakharev, 1981). Their HVR is adapted to optimize SaO_2 and minimize hypocapnia (Santolaya et al., 1989; Rodas et al., 1998). More specifically, Rodas et al. (1998) established that at sea level, Sherpas hyperventilate more with moderate workloads and ventilate less with maximum workloads in comparison to European mountain climbers. If this pattern is maintained at extreme altitude, Sherpas would attain a greater SaO_2 during the habitual exercise in mountain climbing thus avoiding brain vasoconstriction and probably brain dysfunction in intense physical effort. This possibility was explored by Garrido et al. (1996) who studied a group of 7 top Sherpas and a group of 21 top mountain climbers living in lowlands. Both groups did not differ in age or time of exposure to altitudes above 7,000–8,000 m. An extended medical record was made of the subjects about neurological, emotional, perceptive, and neuropsychological symptoms, and they were explored by magnetic resonance imaging (MRI). The results indicated that lowland climbers showed neurological symptoms during ascents (i.e., headaches, ataxia, insomnia, aphasia, vision loss, irritability, hallucinations), while half of them reported neuropsychological difficulties after the descent (i.e., memory impairment, lesser cognitive efficiency, depression). Sixty-one percent of them revealed abnormalities in MRI. Unexpectedly, only one of the Sherpas (14%)

showed alterations in MRI and indicated to have suffered neurological symptoms. Summing up, the experience of altitude does not seem to be an important factor in adaptation as the biological conditioning. The final section of this article considers possible training factors. On the other hand, interaction between effort and altitude is observed in the levels of hypoxia although such interaction has not been checked considering ANI a second dependent variable.

The adaptation of organisms to high altitude should be approached from a broad perspective. So together with the relevant respiratory factors one should consider compensatory vascular mechanisms.

VASCULAR AND HEMATOLOGICAL ADAPTATION ASSOCIATED WITH ALTITUDE

A posterior adaptive process that takes place in longer stays, more than a week at high altitude is polycythemia. Polycythemia provokes an increase in hematocrit due to erythropoiesis mediated by erythropoietin. This process rises the arterial oxygen uptake, which reduces PaO_2 due to a reduced plasmatic pressure of free oxygen (Richalet et al., 1994). This facilitates the gas exchange in the alveolar–arterial membrane. At brain level, the reduced cerebral perfusion is balanced by a response of compensatory vasodilatation that maintains CBF constant. Normality in CBF by means of this strategy can be kept up to levels of 60 or 70 mmHg PaO_2 (i.e., 3,500 m approximately). Given the limit of vasodilatation, beyond this point, the organism compensated the oxygen deficit by increasing CBF (Baumgartner et al., 1999; Jensen et al., 1990). For instance, the reduction of PaO_2 to 25 mmHg produces an increase of 400% in CBF, although obviously at these levels the increase of CBF, obviously, loses capacity to keep constant the oxygenation of the neural tissue (Adams et al., 1999).

On the other hand, the drop in PAO_2 produces a lung vasoconstriction mediated by endothelin 1 produced by endothelial cells (Morganti et al., 1994). Lung arterial hypertension generates a hemodynamic overload in the right ventricle with a clear effect in the increase of electroencephalographic activity (Karliner et al., 1985; Milledge, 1963; Peñazola and Echevarría, 1957). This sequence of events may lead in, extreme cases, to lung edema (West and Mathieu-Costello, 1992). As Moore et al. (1992) have pointed out, this process, observed in Andean communities, does not appear among Sherpas and Butanese, perhaps because these populations have inhabited their regions for 25,000 years. On the contrary, Andean communities were established 10,000 years ago. That much

longer period allows a better adaptation to the hypoxic environment as postulated by Moore et al. (1992).

SLEEP ALTERATION AND NEUROPSYCHOLOGICAL FUNCTIONING

The first research to make a record of the physiology of sleep at altitude was performed by Reite et al. (1975). In this study, six subjects were taken to an altitude of 4,300 m where they underwent a polysomnographic record during 4 nonconsecutive nights. Previously, two nights had been recorded at sea level. It should be stressed that the subjects had not made any special exercise during the stay at altitude, and that they were taken there by helicopter. The most remarkable results were the increase in sleep phases III and IV, and a hardly significant tendency to reducing rapid eye-movement sleep. In addition, the total amount of sleep decreased but in a limited proportion compared with the subjective sensation of not having slept mentioned by the subjects. Another effect was the increase in arousals during the night (19.5 in normoxia against 29.2 at altitude). A new finding was the appearance of PB that was observed in some subjects during most of the night, especially during sleep phases of slow waves. The results showed that arousals were more likely during periods of PB coinciding with the transition from apnea to the beginning of PB. However, there was no relationship between the number of arousals and the time spent in PB. An extreme case was subject 1, who did not show PB at all but displayed the most outstanding increase in arousals after exposure to altitude.

Anholm et al. (1992) placed five subjects under a simulated pressure of 282 mmHg for 6 weeks making polysomnographic records. His results are similar to those by Reite et al. (1975) but showed a much clearer reduction in rapid-eye-movement (17.9% vs. 4%) and a more significant reduction in the total amount of sleep (337 min. vs. 167 min.). Anholm et al. (1992) specified that at 282 mmHg, the mean SaO₂ during the night was 52%. Subsequent investigations have revealed that both PB and sleep apnea during exposure to altitude are related to ventilatory sensitivity to hypoxia (Lahiri et al., 1983; Lahiri and Data, 1992). Specifically, a higher HVR entails a more extended PB during the night. Others have put forward the possible explicatory role of PB in the neuropsychological impairment of mountain climbers (Hornbein et al., 1989; Hornbein, 1992). In this research a more considerable neuropsychological impairment was found after a simulated ascent to 8,848 m in subjects who had displayed a greater HVR, and who paradoxically show a better physical adaptation to altitude established by

number of symptoms of AMS (Masuyama et al., 1986; West, 1984). Likewise, subjects who showed a high HVR maintained their PaO₂ above 30 mmHg (above 8,000 m). Below this point it has been observed that hypoxia-related alterations suffer a qualitative change (Schoene, 2001) as a result of an alteration in the gradient of alveolar pressure that prevents gas transfer.

As mentioned above, it has been suggested that the periods of hypoxemia suffered during sleep time with PB associated with a high HVR were responsible for a greater cognitive impairment at altitude. This hypothesis, however, does not hold since more recent evidence has demonstrated that subjects with a reduced HVR also show periods of acute hypoxemia during sleep that are not related to periods of PB (Coote et al., 1993a; Hackett et al., 1987; Sutton et al., 1987). Matsuzawa et al. (1994) recorded electroencephalographic and electrooculographic activity, and SaO₂ of a group of nine subjects during the first night of sleep at a simulated altitude of 3,700 m. It was observed that saturation fell by a mean of 7% during the night ($p < .01$), the symptoms of AMS were stronger the morning after, and eight of the subjects showed PB between 0 and 57.8% of the night. No correlation was found between PB and SaO₂, PB and the symptoms of AMS, nor between PB and HVR.

Furthermore, it has been noticed that arousals during the night protect against severe deprivation of oxygen. In his pioneering research Reite et al. (1975) did not find connection between apneas or PB during sleep and number of arousals. More recently, Khoo et al. (1995), in research with nine subjects inside a hypobaric chamber using a range of altitudes between 4,500 and 7,500 m noticed a clear association between number of apneas and number of arousals. This supplies further evidence that subjects with a high HVR, and consequently more frequent PB and apneas during sleep, do not suffer more acute night hypoxia. In the results obtained by Khoo et al. (1995), the number of apneas only explained a 20% of the arousals so that arousals should be taken as an independent protective mechanism against a hypoxic environment, as suggested by other authors (Coote et al., 1993a; Coote et al., 1993b; Selvamurthy et al., 1986).

Another way of elucidating the possibility that night hypoxia is responsible for the impairment would be to check whether the administration of O₂ during sleep, which has a demonstrated effect on PB (Reite et al., 1975), reverses the pattern of impairment in subjects with a high/low HVR. Future research should test this hypothesis.

Until new studies shed light on this question, one can rely on a clinical model of chronic hypoxia such as OSA, whose pattern of associated neuropsychological

impairment is generally similar to that of exposure to altitude as explained below (Grant et al., 1987; Kelly et al., 1990; Telakivi et al., 1993). Nevertheless, it should be remembered that this is a case of chronic, not acute, hypoxia. Findley et al. (1986) found that in subjects with OSA who suffered associated hypoxemia, the total degree of neuropsychological impairment correlated with the levels of PaO_2 and SaO_2 both measured during the night or the daytime. Unfortunately, these results only indicate that hypoxia related to altitude during sleep could be enough to explain the associated neuropsychological impairment. However, at altitude, oxygen desaturation is still present in the daytime and the periods of hypoxia are shorter than those suffered by patients with chronic OSA.

In conclusion, during sleep at high altitude there is a reduction in the later sleep phases of slow waves and an increase in phase 1, which produces impairment in the quality of sleep (Matsuzawa et al., 1994; Wickramasinghe and Anholm, 1999). Rapid eye-movement sleep is reduced and night arousals increase significantly (Buguet et al., 1994). Finally, there is a considerable drop in oxygen saturation values during the night, a phenomenon not clearly associated with AMS, PB, or HVR that can contribute strongly to neuropsychological impairment. Interestingly, this pattern of alterations is hardly observed in Andean communities living at high altitude (Coote et al., 1992; Coote et al., 1993b).

ACUTE MOUNTAIN SICKNESS: SUFFICIENT BUT NOT NECESSARY CONDITION FOR ALTITUDE NEUROPSYCHOLOGICAL IMPAIRMENT

AMS should be distinguished from concurrent neuropsychological or behavioral alterations. AMS is associated with too sudden an ascent for the body to perform compensatory actions, especially ventilatory and vascular, for a reduced environmental oxygen pressure (PO_2), while psychological alterations are related to the consequences of a reduced PaO_2 in the organism independent of an adequate or inadequate acclimatization. In other words, it is not necessary to have AMS to detect neuropsychological alterations since compensatory mechanisms do not neutralize completely the effects of hypoxia on the nervous system and consequently on behavior. For example, Kramer et al. (1993) did not find any correlation between the degree of AMS and the increase in latency in RT. Neither did they connect AMS and the degree of dysfunction in a battery of perceptive, cognitive, and motor tests. In this case, subjects were exposed to a moderate rate of ascent so that AMS was prevented (none of them surpassed

a score of 3 out of a 6-point scale of AMS). However, this did not avoid the presence of a set of psychological consequences, despite a mean ascent of 300 m per day favored an adequate physical adaptation. Also, Krammar et al. (1983) did not find any relationship between altitude and AMS at a similar rate of ascent. For this reason, it is necessary that studies not only specify altitude but also rate of ascent. Above mean values used by Krammar et al. (1983) there is indeed a positive association between altitude and symptoms of AMS (Nicolas et al., 1999). Considering that both altitude and the pace of ascent are associated with the presence of AMS and neuropsychological impairment (Hansen et al., 1967; Shukitt and Banderet, 1988), *a priori* or *a posteriori* controls should be made of the symptoms of AMS in order to determine which effects are the result of a normal adaptation to altitude and which correspond to this disorder. Studies made about high rates of ascent have found that the pattern of effects may differ whether the subject develops AMS or not. Regard et al. (1991) have established a double dissociation between subjects who developed AMS and subjects who did not. A fast ascent to an altitude of 4,500 m was made inside a hypobaric chamber. The ascent took 24 hours. This rate avoided the ground effect that similar investigations have suffered in trying to associate the symptoms of AMS with the psychological effects of altitude (Kramer et al., 1993). Subjects who developed AMS showed little deficit in short-term memory and a more significant impairment in conceptual tasks. On the contrary, those who did not develop AMS displayed the reverse pattern: a greater impairment of short-term memory and scarce alteration in conceptual tasks. Similar findings were obtained by Forster (1985).

It should be stressed that a human being who is still conscious with a PaO_2 of 30 mmHg (equivalent to an altitude higher than 8,000 m) after an adequate rate of ascent must have an enormous capacity of acclimatization. A chronic hypoxia model of this phenomenon can be found in COPD, who live normally around the threshold of 30 mmHg of PaO_2 (Adams et al., 1999). In contrast, a nonacclimatized subject exposed to such conditions would die or lose consciousness, as was dramatically observed in balloon ascents in the 19th century (Bert, 1878/1978). The case of Gaston Tissandier in 1,875 made the scientific community aware of the mortal danger of too fast ascents. Tissandier and the brothers Croce-Spinelli ascended together in the balloon *Le Zénith* to an altitude of 8,600 m. Above 8,000 m they all lost consciousness. Tissandier brightened up after the descent. The others perished (Tissandier, 1875).

After considering the respiratory and vascular effects associated with acute hypoxia and being aware that the causal factors of neuropsychological impairment are

hypoxia and cerebral ischemia, we are going to analyze the nature of the brain dysfunction related to altitude.

BRAIN DYSFUNCTION ASSOCIATED WITH ALTITUDE: STRUCTURAL AND FUNCTIONAL ALTERATIONS

Despite vasodilatation, polycythemia, and the increase in CBF associated with hypoxia, the human organism cannot avoid a drop in oxygen saturation in the central nervous system (CNS) in conditions of high or extreme altitude. Desaturation is particularly intense if the subject exercises (Bradwell et al., 1999; Roach et al., 2000; Saito et al., 1999; Sutton et al., 1988), sleeps at altitude regardless of the presence of PB (Coote et al., 1993a; Hackett et al., 1987; Matsuzawa et al., 1994) and shows a high hypocapnia associated with an intense HVR (Hornbein et al., 1989; Schoene, 1999). Brain desaturation resulting from these phenomena may explain reversible neuropsychological alterations associated with altitude. Such processes are independent of AMS, whose symptoms display a different temporal pattern (Shukitt-Hale et al., 1991). In addition, the symptoms of AMS are independent of the occurrence of neuropsychological impairment (Bahrke and Shukitt-Hale, 1993) although AMS entails qualitative changes in the type of dysfunction, as discussed above.

Blood hyperviscosity due to the increase of hematocrit (Samaja et al., 1993), alterations in coagulation (Doughty and Bearmore, 1994), and probably the increase in intracranial pressure (Roach and Hackett, 2001) augment the probability of thrombosis during prolonged stays at high and extreme altitude. These alterations may correspond to focal ischemic infarcts in the CNS (Garrido et al., 1993; Garrido et al., 1995; Song et al., 1996) that develop with acute neurological symptoms from mild to severe. Among others, ataxia (Hackett et al., 1998), aphasia (Garrido and Javierre, 1996), blindness (Hackett et al., 1987), hemianopsia (Garrido et al., 1993), diplopia (Shlim et al., 1995), hemiparesis (Basnyat, 1997), delirium (Basnyat, 2002) and global amnesia (Litch and Bishop, 1999), have been described in this context. Ischemic accidents together with a sufficiently extreme and prolonged desaturation to interrupt the neuronal metabolism and produce necrosis are probably the most relevant factors to produce short and long-term ANI after returning to lowland. The possibility of irreversible brain dysfunction due to hypoxia in absence of ischemia remains a matter of debate (Hornbein, 2001; Simon, 1995).

Hypoxia damages all necessary aerobic processes maintaining Krebs cycle and the electronic transport sys-

tem. When neurons are deprived of oxygen, they catabolize themselves. The resulting accumulation of catabolic products such as lactic acid causes irreversible damages that eventually result in the death of the cell (Adams et al., 1999). Besides this mechanism, in the case of glutamatergic neurons, hypoxia induces an excessive release of glutamate. Once it is linked to *n*-methyl-D-aspartatus (NMDA), this neurotransmitter generates an excessive flow of sodium ($[Na^+]_e$) and calcium ions ($[Ca^{2+}]_e$) in the postsynaptic neuron. This causes cellular death after the postsynaptic neuron has released its reserves of glutamate producing a toxic flow that extends to adjacent neurons (Schousboe et al., 1997; Pinel, 2001). The considerable presence of glutamatergic cells in the hippocampus explains partially the enhanced sensitivity of this structure to conditions of hypobaric hypoxia (Gozal et al., 2001; Shukitt-Hale et al., 1996). These findings have not been confirmed in human investigations analyzing through imaging techniques the damage to the encephalon after climbs to peaks above 8,000 m without supplementary oxygen (Garrido et al., 1995; Garrido et al., 1996; Hackett et al., 1998).

Structures Sensitive To Hypoxia

The direct assessment of brain damage associated with altitude is limited so we have turned to considering laboratory findings related to other hypoxic accidents. Brierley (1976) observed that in hypoxic conditions, the most damaged structures were the hippocampus, the thalamus, layers 2, 4, and 5 in the cortex, the amygdala, and the corpus striatum. Pichiule et al. (1996) obtained similar results. These authors exposed rats to intermittent hypoxia (4,300 m) during 10 weeks. They observed a reduction in the agonist ligament (2H)MK-801 of the NMDA receptor with respect to 36% control in the case of the cortex, 35% in the hippocampus, and 31% in the striatum. Reed et al. (1999) observed by MRI that four subjects who had suffered severe hypoxia and showed anterograde amnesia had a clear damage in the middle temporal lobe comprising all the hippocampus and the parahippocampal areas. Adams et al. (1999) observed that subjects with hypoxic encephalopathy of various etiologies showed damages in inferomesial areas in both temporal lobes, and problems of memory and anterograde amnesia in accordance with altitude. Schulze et al. (1990) observed that exposing his subjects to a SAO_2 between 88% and 90% (equivalent to an altitude of 2,500 m) produced a metabolic delay in the hippocampus, hypothalamus, cortex, and striatum. The reduction in the release of acetylcholine (Ach) in rats exposed to 5,500 m was studied by Shukitt-Hale et al.

(1993). The physiological measurement was more sensitive than the behavioral assessment, in this case the Morris Water Maze (MWM). This test did not indicate significant damages up to 5,950 m. This author noticed damages in area CA3 in the hippocampus after exposing animals to an altitude of 6,400 m for 4 days (Shukitt-Hale et al., 1996). Gozal et al. (2001) used immunohistochemistry to record an increase in apoptosis in area CA1 of the hippocampus after 2 days of exposure to intermittent hypoxia (10/21% O₂). This alteration reverted after 14 days. Intermittent hypoxia increased latencies and longitude of the paths in the MWM.

Zola-Morgan et al. (1986) have verified in humans the special sensitivity to hypoxia of the hippocampus and the limbic system. In this regard, hypoxia has been significantly associated to tests that measure dysfunctions to the hippocampus such as the MWM (Shukitt-Hale et al., 1996). Prigatano et al. (1983) and Bedard et al. (1991) maintain that the frontal lobe is also affected in subjects who suffer from hypoxia continuously. In support of this hypothesis, Viapiano et al. (2001), exposed a group of 21 monkeys to hypoxia, found a 30% decrease in maximal binding capacity in γ -aminobutyric acid (GABA) to GABA_A receptor sites. The change was reversible after returning to normoxia. Adams et al. (1999) point out that the hippocampus and the deep cerebellar folia are the most sensitive structures to anoxia.

Specific investigations by Regard et al. (1989) found that two subjects from a group of eight world-class mountain climbers who had ascended repeatedly above 8,000 m without supplementary oxygen experienced chronic anomalies in their temporal and frontal electroencephalographic activity measured months after their last ascent.

NEUROIMAGING RESEARCH

Using MRI techniques, Anooshiravani et al. (1999) did not detect functional nor structural alterations in a group of climbers after returning from a 6,000 m peak. On the contrary, Garrido and his team have found on three occasions structural alterations in the brain through MRI in subjects who had ascended above 8,000 m without supplementary oxygen (Garrido et al., 1993; Garrido et al., 1995; Garrido et al., 1996). In the first case, in a sample of 26 mountaineers who had climbed to peaks above 7,000 m they detected anomalies in MRI with respect to controls. The authors observed anatomical anomalies in 46% of the subjects, basically signs of cortical atrophy in five subjects, and leuko-araiosis in five other subjects. Two other subjects displayed both alterations in MRI. It should be noted that this research has a transversal design

not a longitudinal one. A control group of nonclimbers was selected and the neuroimaging records were made up to 3 years after the ascent. Another study by the same authors (Garrido et al., 1995) without the said shortcomings obtained similar results. This time, after an ascent, alterations in the form of high intensity signals were detected in 56% of a sample of nine subjects (five subjects) in periventricular areas (leuko-araiosis), posterior parietal cortex, and occipital white matter, thus confirming previous observations (Griggs and Sutton, 1992). In a third study, the authors replicated these findings (Garrido et al., 1996) with a group of 21 lowland elite mountaineers with an average stay of 445 hours above 7,000 m and 56.5 hours above 8,000 m. In this case, 61% of the subjects showed signs of cortical atrophy and high-density signals while no abnormalities were reported in the controls. Alterations were mainly found in the atrial zone and the periventricular regions of the posterior horns of the lateral ventricles. In this latter study, these results did not appear when a third group of seven world-class Sherpas was evaluated. On the other hand, no associations were found between anomalies obtained by MRI and neuropsychological alterations described by medical history and neurological examination (no formal neuropsychological assessment was arranged). The number of ascents, time of exposure to extreme altitude, and time elapsed since previous ascents were not correlated with MRI findings. Hackett et al. (1998) examined nine subjects with High-Altitude Cerebral Edema and noted that seven of them showed intense signals in T2 in different areas of the white matter, especially the splenium and the corpus callosum.

Leuko-araiosis observed by means of MRI is associated with cortico-subcortical injury with a widening of Virchow-Robin spaces that contain the perforating vessels of the middle cerebral artery. Histologically, leuko-araiosis shows demyelination, axonal degeneration and astrogliosis considering a partial infarct of the white periventricular matter. Leuko-araiosis is a likely consequence of intermittent anoxias due to alterations in the regulation of the cerebral flow in long and thin vessels with no anastomosis (Garrido, 1997). In the absence of ventricular enlargement, leuko-araiosis is associated with a slow-down in complex mental processes (Junqué et al., 1990) that can be similar to that frequently suffered by climbers at extreme altitude and identified by Ryn (1988) as *acute organic brain syndrome*. A significant association has been reported between leuko-araiosis and reduced speed in complex information processing and with initial dementia in elderly people (Steingart et al., 1987; Junqué et al., 1990). However, the long-term evolution of such neuroimaging findings observed in world-class Caucasian climbers is not known (Garrido et al., 1996).

In this context, considering the oxygen uptake previous to 8,000 m, the negative result obtained by Anooshiravani et al. (1999) seems logical, rather than contradictory, since their subjects did not go above 6,000 m and were exposed to a shorter stay at altitude than those studied by Garrido and his team. However, Garrido et al. (1995) stress that 80% of their subjects who showed alterations in neuroimaging before the ascent had never climbed higher than 6,800 m before the studied expedition. This contradicts Anooshiravani's results as long as it is assumed that the dysfunctions observed in these subjects were due to hypoxia suffered in previous expeditions.

Recent techniques to record brain metabolism *in vivo* have shed light into this issue. Hochachka et al. (1999) evaluated the regional metabolism of the brain glucose in six subjects by means of Positron Emission Tomography (PET) before and after training during 63 days at an altitude between 3,181 and 6,157 m. These authors found a reduction in the brain activity in three frontal areas, the left occipital lobe, and the right thalamus. A bilateral increase in the cerebellum activity was also observed. The effects were intense with oscillations between 10% and 18% of the absolute metabolism compared with the first measurement. These striking results were not be replicated by Moller et al. (2002). These researchers established global brain metabolic rates for oxygen and glucose, oxygen delivery, and CBF before and after a stay at altitude of a group of nine lowlanders, without finding differences between both conditions neither in exercise nor at rest. The difference with findings by Hochachka et al. (1999) is due to their global, not regional, measurements, their different techniques (Kety-Schmidt vs. PET), a shorter exposure (21 vs. 65 days), and a lower maximum altitude (5,260 m vs. 6,157 m). The alterations observed by Hochachka et al. (1999) are not found among Sherpas, who have a brain metabolism at altitude similar to that of lowlanders in normal conditions, as measured by PET (Hochachka et al., 1996). Unfortunately, these studies did not include measurements of neuropsychological functioning.

Extreme desaturation, together with the presence of focal ischemic processes provoked by hypoxia produce neuronal death in mountaineers exposed to high and extreme altitude extendedly or repeatedly. MRI has shown focal ischemic damages in periventricular areas and cortical atrophy in climbers exposed to extreme altitude. These findings have not been replicated in exposure to moderate and high altitude. It is hypothesized that a mild damage can happen in structures sensitive to hypoxia, such as the hippocampus, due to a high desaturation. This has been tested in animals but not in humans. The new application of functional neuroimaging techniques has demonstrated the

regional, but not global, reduction of the CBF especially in frontal areas. Because most of the neuropsychological impairment associated with altitude is reversible, and the circumstance that no neuropsychological measurements have accompanied functional records of brain activity, it is not possible to reach conclusions about physiological correlates at CNS of ANI. Future research using PET or functional MRI, together with an extensive neuropsychological record will improve the knowledge of brain function underlying ANI.

PSYCHOMOTOR EFFECTS OF ALTITUDE

A most debated effect of altitude has been the impairment of motor performance. It was already pointed out by McFarland (1937a, 1937c), and has been measured by most recent investigations on this issue. The effect manifests as a lesser motor speed and precision as compared to subjects' sea level performance (Berry et al., 1989; Bolmont et al., 2000; Hornbein et al., 1989; Sharma et al., 1975; West, 1984). Differences in RT cannot be attributed to this effect, as argued below.

The most widespread test to detect motor speed impairment is the FTT from the Halstead-Retain Neuropsychological Battery. This test has proved its efficiency to discover slight motor dysfunctions (e.g., Peña-Casanova et al., 1997). Other motor tests have obtained similar effects, for instance the Purdue Pegboard Test, sensitive to speed, motor coordination and precision (Bolmont et al., 2000; Lezak, 1995).

It is debatable whether motor effects are the direct result of hypoxia or due to related factors. In this respect, investigations have highlighted the modulating role of anxiety and fatigue in connection with motor performance in experiments of exposure to simulated altitude. Several reports from Operation Everest II and Everest-Comex 97 supply some relevant results. We argue that the hypothesis that psychomotor effects are an indirect result of exposure to altitude is supported by three types of observations. First, experimental data indicate the absence of neuromuscular alterations at simulated altitude up to 8,848 m. This is consistent with the possible effect that fatigue is a central, rather than peripheral symptom (Garner et al., 1990). These authors measured 3 torques during a 40-day simulated ascent to 8,848 m in a hypobaric chamber (Operation Everest II Project), muscle twitch torque, titanic torque and maximal voluntary contraction torque. Although the amplitude of muscle activity was not affected at 6,500 m and 8,848 m conditions, they found evidence of fatigability of the twitch in a test interpolating a 20 Hz stimulus into the muscle. This result accords with the

emergence of central fatigue (Garner et al., 1990). Second, psychophysical findings demonstrate that alterations in RT do not obey motor commitments (Abraini et al., 1998; Bouquet et al., 1999). These studies have shown that performance in psychomotor activity in an altitude-dependent manner considering altitudes above 6,000 m (Everest-Comex 97 Project), while RT remains at basal levels in all altitude conditions. These data shows that basic motor processes remain unaffected. Third, it has been noted that fatigue established by the Profile of Mood States is significantly associated with performance in the Purdue Pegboard Test, but not with RT, assigning new value to the central mediation hypothesis (Bolmont et al., 2000). Conversely, Hornbein et al. (1989) as part of Operation Everest II Project reported that the poorer performance in the FTT was not significant in a group of simulated ascent to 8,848 m, whereas in a group that climbed to Mount Everest there was a significant impairment ($p < .001$). These paradoxical results as compared with the Everest-Comex 97 data call for further investigation in order to properly establish the particular conditions in which psychomotor impairment can be replicated *in vivo* and *in vitro*.

Possible Permanent Dysfunctions

On the contrary, a second group of observations indicates that motor impairment is a genuine effect of altitude. The long-term permanence of motor impairment after returning to sea level supports the hypothesis that this alteration cannot be an indirect consequence of altitude by the effect of anxiety or fatigue. During the American Medical Research Everest Expedition (AMREE-1981), Townes et al. (1984) and West (1984) noticed that 15 out of 16 mountaineers showed a significant impairment in the FTT right after the expedition and that 13 of them still showed it a year later. This expedition did not use supplementary oxygen. West (1986) confirmed that impairment was still significant 2 years later. These results should be taken cautiously because this investigation did not take into account whether the climbers were exposed again to extreme altitude. Then again, Regard et al. (1989) gave the same test (FTT) to world-class climbers who had not been exposed again to altitudes above 5,000 m during an average period of 7 months. Their results showed that the motor impairment was significant in two out of the eight subjects examined with respect to controls.

Neither hypothesis is conclusive if one considers that both hypoxia and fatigue are modulated by the workload. A higher degree of fatigue is likely to increase the levels

of hypoxia and produce long-term effects, though this remains an object for future investigations.

The finding of permanent effects of altitude depends on three factors, only one of which is irreversible and actually permanent. (a) There may be a diffuse brain damage in areas especially sensitive to hypoxia through repeated and prolonged exposure to altitudes above 8,000 m. (b) Subjects may have been repeatedly exposed to altitude during the posttest period. (c) The posttest may be performed during a period of 45 days after the descent. This means that subjects have an increased HVR and accordingly one cannot be sure that they have returned to normoxia (Forster et al., 1971; Masuyama et al., 1986). Factors (b) and (c) should be taken into account because the available evidence indicates that, although effects can be long-lasting, they tend to disappear in less than 2 years (Townes et al., 1984; West, 1986). Alternatively, when no such long-lasting effects have been found the reasons seem to be the types of tests used, durations of exposure (i.e., shorter durations), maximum altitude reached (i.e., lesser altitude), the use of supplementary oxygen, or an inadequate timing of the posttest measurement (i.e., several days or weeks after the descent). These factors partly explain the negative results of some investigations (e.g., Clark et al., 1983; Jason et al., 1989; Milne and Gray, 1983).

In order to justify the genuine effects of altitude on motor performance we can observe the convergence of results among clinical models of chronic hypoxia. Bedard et al. (1991) noticed that in subjects with OSA, a poorer performance in the Purdue Pegboard Test was associated with higher levels of apnea. McSweeney et al. (1985) obtained similar findings among subjects with COPD using the FTT. However, results in this area are inconclusive (Grant et al., 1987; Stuss et al., 1997). A third element that supports the genuine effect of altitude is the finding of psychomotor impairment in laboratory experiments for controlling fatigue and anxiety effects (e.g., Berry et al., 1989). In addition, experimental research has shown that a poorer performance in the FTT is strongly related to the reduction in oxygen saturation in a higher proportion than its correlation with anxiety. Berry et al. (1989) detected a linear tendency between oxygen saturation (between 100% and 80% SaO₂) and the number of tappings ($p = .009$). In a laboratory study, Hornbein et al. (1989) established a correlation of -0.98 ($p < .05$) with PaO₂, and 0.85 ($p < .05$) with SaO₂ using the same test. Additional evidence for this hypothesis is supplied by the permanence of effects after prolonged exposure to altitude. Sharma et al. (1975) recorded motor speed and efficiency at different moments in a community that stayed at 4,000 m for 2 years. Psychomotor efficiency suffered an

initial impairment and disappeared several months after the beginning of the stay. The speed variable, strongly associated with the task of the FTT, remained altered throughout the stay at altitude, too long to be considered an effect of anxiety, fatigue, or novelty.

At etiological level, psychomotor impairment seems to be the result of a reduction in oxygen saturation since hypocapnia associated with acute hypoxia does not alter the pattern of motor effects when the levels of CO₂ (isocapnia) are stable, as observed by Berry et al. (1989). Unfortunately, these authors did not apply sufficient oxygen saturation to establish the vasoconstrictor effect of hypocapnia. The sensitivity to conditions of hypoxia of motor structures such as the basal ganglia or the cerebellum (Adams et al., 1999) can be a decisive factor to display such effects.

The minimum height that produces motor impairment varies among investigations between 2,500 m and 6,000 m. Some consensus will necessarily be reached when similar tasks are used. For instance, Bolmont et al. (2000) has not found significant dysfunctions below 8,000 m using the Purdue Pegboard Test, while Berry et al. (1989) observed them at a mean altitude of 3,500 m using the FTT.

Performances in the FTT and HVR level were not correlated, detaching psychomotor impairment from other cognitive processes whose levels were significantly correlated with HVR in the report (Hornbein et al., 1989). Psychomotor impairment process may follow a specific temporal pattern of development or simply indicate a flaw of this investigation to detect a significant connection. In this respect, FTT results show a high correlation with SaO₂ ($r = .85$; $p < .05$) for a simulated altitude of 8,100 m and 8,848 m, respectively. Furthermore, the effect of HVR on the %SaO₂ takes only a few seconds of latency (Khoo et al., 1995).

Regarding clinical models of chronic hypoxia, alterations are not convergent. In the case of COPD, simple psychomotor skills, and psychomotor speed are resilient to hypoxia. Motor alterations are only detected in subjects suffering from severe hypoxia (Grant et al., 1987).

We have argued that psychomotor effects seem a direct consequence of hypoxia, as suggested by its association with biological markers of hypoxia and its permanence after return to lowlands. The effect is modulated by anxiety and fatigue. In the case of fatigue, it is presumed to have an indirect effect increasing the degree of hypoxia. In contrast, impaired speed is detected at lower altitudes than impaired precision. It is advisable for future research that speed and precision tests are applied concurrently. For instance, using both the FTT and the Grooved Pegboard Test, a test more focused on precision and coordination.

EFFECTS OF ALTITUDE ON PERCEPTION

Basic Processes: Reaction Time

The effect that meets most acceptance in the literature on the effects of altitude is the increase in complex reaction time (CRT). It has been found both in laboratory conditions (Bolmont et al., 2001; Fowler et al., 1987) and real expeditions (Kramer et al., 1993), and both with and without AMS (Mackintosh et al., 1988), although AMS triggers a greater CRT increase compared to baseline levels (i.e., low-altitude, sea level P_B). CRT is a sensitive index of exposure to altitude although most effects do not appear up to 6,000 m and after prolonged exposure (Hornbein et al., 1989; West, 1984), significant increase of CRT can be found above 2,500 m. Denison et al. (1996) found increases in CRT at such low altitude as 1,500 m. Fowler et al. (1987) demonstrated that increase in CRT in their subjects was significant from 2,438 m upwards, considering altitude by means of induced %SaO₂.

A better understanding of the factors that affect this process has been gained by using event-related potentials (ERP). Wesensten et al. (1993) recorded the auditory CRT of their subjects using a hypobaric chamber to induce a hypoxia corresponding to 4,300 m. They observed that the increase in CRT due to hypobaric environment corresponded to increases in latency of component P300. P300 also suffered a reduction following a temporal pattern that differed from the increase in latency and CTR, which suggests two different mechanisms. In contrast, no effects of altitude were found in components N100, P200, and N200. Considering the amplitude of P300, Fowler and Prlic (1995) observed that it displayed an inverted-u relationship with the degree of hypoxia, while the increase in CRT and P300 latency showed a linear co-variation. This provided additional evidence of the existence of two different mechanisms. More recent research has confirmed the findings related to P300 and demonstrated increases in latency of N200 (Kida, 1997).

Fowler and Lindeis (1992) applied the oddball paradigm (Carretié and Iglesias, 1995) to the auditory ERP and also found an increase in latency of the component P300. These authors obtained the same results in visual ERP. The literature on the topic coincides in the attribution of the increase in CRT to the rise in latency of P300 setting aside motor factors in this case (Fowler and Adams, 1993). This evidence can be interpreted as a slowing in the identification of stimuli, given the increase of P300 latency typically related to this function (e.g., Johnson, 1988), which results in an increased CRT. Fowler and Prlic (1995) measured auditory CRT and ERP at different levels of saturation between 86% and 77%

SaO₂ observing equivalent linear relations between desaturation and CRT increase, and between desaturation and P300 latency. However, Fowler and Adams (1993) found that previous levels of processing can also be affected. According to their results using the method of additive factors, below a level of oxygen desaturation between 63% and 64% they found a high interaction between degree of hypoxia and stimulus intensity, and CRT. The effect of the increased stimulus intensity is considered to be associated with a preprocessing stage rather than with a stimulus identification stage.

Takagi and Watanabe (1999) applied the paradigm of contingent negative variation, where a subject has to choose between two answers, whether the stimulus shown was the same as, or different from, another shown a second earlier. This is related to motor programming, attention, motivation, and memory (e.g., McCallum, 1988). The said study found a reduction in contingent negative variation as an effect of a 6,000 m hypoxia especially in earlier components to the detriment of later components. Conversely, up to 3,000 m the amplitude of later contingent negative variation was negatively correlated with CRT while above 3,000 m it was the amplitude of the early component that correlated with CRT negatively.

Basic Processes: Perceptive Discrimination

Watson et al. (2000) observed that ERP and increase of CRT were not associated with a rise in the absolute threshold for auditory stimuli up to 16 kHz at a hypoxia equivalent to altitudes of 1,200, 2,400, and 3,700 m. Fowler and Grant (2000) obtained similar results. Burkett and Perrin (1976) noticed that at 6,600 m altitude had no effect on the discrimination of speech sounds. Finally, in an analogous investigation, Martin et al. (2000) found no effects on the localization of stimuli at an altitude of 3,700 m.

The findings in discrimination of auditory stimuli cannot be extended to other sensory modalities. Typical findings in exposure to altitude include reductions in the absolute threshold for tact, detection of CO₂, smell, light, and taste (Fleisch and Von Murant, 1944, 1948; Kobrick, 1983). The same alteration in threshold was observed in the sensation of breathing (Nöel-Jorand and Burnet, 1996). These reductions vary between 25% and 40%, and are clearly associated with acute effects of hypoxia, since administration of oxygen reverts the effects quickly. Nöel-Jorand et al. (1996) applied a method of constant stimuli with nociceptive stimuli. They made evaluations before an expedition, during the expedition at 3,500 m and 5,600 m and after the expedition. Clark's Situational Pain

Questionnaire was used. The climbers showed a greater subjective tolerance to pain especially above 5,000 m. Nevertheless, the absolute threshold was reduced coinciding with findings by Fleisch and Von Morant (1944, 1948) for other sensory modalities, that is, climbers had a greater subjective tolerance to pain but perceived painful stimuli of lesser intensity. This effect cannot be attributed to a tendency of response or the consequence of cold, which was also suffered by control subjects and is known to have the reverse effect (Plaghki et al., 1994). In addition, the presumed stoicism of mountaineers should be distinguished from sport induced analgesia (Padawer and Levine, 1992) since no analgesia is produced but only an increase in the subjective representation of tolerance to pain (no direct test of pain tolerance was carried out). This result is attributed to adaptation to a situation of precarious survival. Similar results have been obtained in Sherpas and Tuaregs (Clark and Clark, 1980). Nöel-Jorand and Burnet (1996) consider that these sensory changes follow a broad response of adaptation to altitude; however, the relationship between these changes and the degree of adaptation to altitude has not been sufficiently evaluated.

Experiments on adaptation to darkness under hypoxia revealed a rise in the threshold of detection of visual stimuli (Kobrick and Appleton, 1971; McFarland and Evans, 1939). In contrast, the threshold of detection of visual contrast under conditions of severe and prolonged hypoxia was not affected, according to the results of the Ginsburg Vistech Test (Kobrick et al., 1988). Alterations in the perception of progressively greater brightness have been observed in simulated altitudes of 3,962 m, 4,572 m, and 5,186 m (Cahoon, 1970). Another effect of altitude on perception is the alteration in color vision although little research has been done in field conditions. By studying curves of adaptation to darkness during exposure to an altitude of 4,300 m for 2 weeks, Kobrick et al. (1984) observed a rise in the absolute threshold of green that remained throughout the whole function of adaptation and the whole stay. With similar conclusions, Richalet et al. (1988) observed alterations in the green-red axis in a field research. In particular, a reduced sensitivity to green in relation to red was recorded in the measures obtained by a portable analoscope up to an altitude of 4,350 m. The same authors (Richalet et al., 1988) replicated successfully these results at 4,800 m during an expedition to the Annapurna (8,091 m). Laboratory research performed in a hypobaric environment obtained similar results (Smith et al., 1976; Vingrys and Garner, 1987). Recently, Leid and Campagne (2001) have criticized those initial results. They argue that by using a technique that primarily evaluates long wave longitude the shades of red can obtain a lower perception due to the measuring procedure.

Using the Lanthony Test of Desaturation D15, Leid and Campagne (2001), took measurements up to 7,000 m during two expeditions to mountains Cho Oyu (8,201 m) and Gasherbrum (8,068 m) checking meteorology, age, physiological state, AMS, fatigue, and time. This study found a subtle impairment of the yellow–blue axis in two out of five subjects, while the rest of the subjects were not impaired. However, the D15 exhibited a weak sensitivity in the red–green axis so that the results obtained by Richalet et al. (1988) cannot be discarded. Using a discrimination task that is not biased towards any color, Bouquet et al. (2000) obtained intermediate results. The subjects employed by Bouquet et al. (2000) in a simulated ascent of 31 days to 8,848 m performed a discrimination task with different hues in the red, blue, and green ranges. They observed slight impairment that was specially significant in red and blue.

Hallucinatory Experiences

In the list of perceptive alterations due to exposition to altitude, we should take into account hallucinatory experiences. In the history of mountain climbing, plenty of anecdotes have been recorded. Rutledge (1934) reported how the mountaineers climbing Mount Everest in 1933 talked about hallucinations or perceptive alterations such as the feeling of being accompanied to the extent of dividing the food supplies with an inexistent companion. McFarland and Barach (1937) observed the frequent appearance of hallucinations in psychoneurotic patients subjected to hypoxia. Ryn (1988) describes a case of complex visual hallucination at 6,900 m. A subject was reported to have seen red snowflakes falling around him. This hallucination lasted 2 days with a variable intensity and only stopped when the subject descended to 5,000 m. More recently, Brugger et al. (1999) interviewed a group of world-class mountaineers to find out that 88.9 % had experienced some kind of hallucination. Most frequent cases were somesthetic illusions, and visual and auditory pseudo-hallucinations. Garrido et al. (2000) interviewed 33 mountaineers who had climbed above 7,500 m at least once. This time, 32.3% declared having experienced hallucinations: somesthetic illusions (especially the sensation of having an imaginary companion behind), visual and auditory hallucinations. With two exceptions, all the episodes occurred above 6,000 m and their duration lasted between two seconds and several hours. In these cases, it is difficult to dissociate the effects of hypoxia or altitude from those associated with stress, sensory deprivation, social deprivation or the experience of danger that have been associated with somesthetic illusions. These per-

ceptive alterations have not been connected with posterior neuropsychological dysfunction or MRI abnormalities (Brugger et al., 1999; Garrido et al., 2000).

Summing up, it can be said that exposure to altitude produces consistent increases in CRT around 3,000 m which can be attributed to a slow-down in the process of identification of the stimulus, apart from extreme cases where the intensity of the stimulus can increase the CRT. This is consistent with occasional findings of increases in latency of N200 associated with the initial processing of stimuli (Carretié and Iglesias, 1995; Kida, 1997). These findings have been obtained at visual and auditory modalities. In contrast, changes in ERP and CRT cannot be attributed to motor problems since the increase in simple RT can be very slight or inexistent in exposure to altitude (Fowler and Lindeis, 1992). In this respect, we notice that recording the preparation potential associated with the temporal pattern of motor responses will supply relevant information, but we have not found related investigations. Additionally, the reduction in the amplitude of P300 seems to indicate some adaptive process independent of the increase of CRT since it displays a linear and temporal pattern unrelated to changes in RT. Also, discrimination and localization should be discarded as factors in the previous findings. Reductions have been observed in the thresholds of detection of smell, light, taste and pain stimuli. A reduced response is detected in adaptation to darkness. Finally, hallucinatory experiences, especially somesthetic, have been reported in subjects ascending above 6,000 m without supplementary oxygen.

MEMORY, LEARNING AND ATTENTION IMPAIRMENT

The high sensitivity to hypoxia of structures such as the hippocampus and the limbic system (Shukitt-Hale et al., 1996; Zola-Morgan et al., 1986) makes it predictable that exposure to altitude will bring about dysfunctions to learning and memory. The effects on memory have been noticed by numerous studies using diverse methods (Bakharev, 1981; Bedard et al., 1991; Kramer et al., 1993; Prigatano et al., 1983; West, 1986). Certain evidence indicates that these memory difficulties depends on a reduced capacity to learn new information rather than its retrieval (Cavaletti et al., 1987; Kennedy et al., 1989; Oelz et al., 1986; Townes et al., 1984).

Retrieval Vs. Storage Deficits

Initially, Shock (1942) demonstrated that a 10% reduction in oxygen concentration (90% SaO₂) interfered

with the learning of a visual discrimination task in rats inside a maze. This time, hypoxia had a stronger effect in acquisition than in retrieval. Using Facretreval2, a test to dissociate memory problems in humans due to difficulties in information retrieval or storage, Nelson et al. (1990) found that up to 6,000 m altitude played no effect on retrieval but learning new information was impaired. However, as suggested by animal research, severe hypoxia is likely to dysfunction the retrieval of known, and even automatized, tasks. In a study with rats, Chleide et al. (1991) found dysfunctions in the retrieval of a learned task after inducing hypoxia for 4 days and allowing 6 days for rest before testing an instrumental conditioning task. The performance of the test on successive days showed that the effects on retrieval were reversible. At any rate, these results should be taken cautiously because researchers used food as reinforcement ignoring the reduction in food craving and consumption in rats under hypoxia (Ettinger and Staddon, 1982). In humans, Kramer et al. (1993) obtained a similar result by observing the effects of altitude (6,100 m) on an actual expedition performing a task that required controlled processing (targets and distractors exchanged their roles during the task), and a second task demanding automatic processing (every stimulus evoked a constant response throughout the experiment). The percentage of transfer between sessions for the RT showed that the automatized task was more impaired than the controlled task in comparison with the control pattern. Control and experimental subjects displayed 80% transfer against 25.5% for the automatic task, and 95.6% against 80% for the controlled task, respectively. The authors observed that dysfunction affected other learning tasks (both with perceptual and semantic material) even 2 days after descending to sea level. In another study, Koller et al. (1991), observed dysfunctions in the execution of a well-learned task (mental calculation) that demands controlled processing during exposure to an altitude of 6,000 m. The effect was stronger if subjects were not acclimatized (20% vs. 7% error).

Nicholson and Wright (1975) observed that under moderate hypoxia (4,572 m, 50 mmHg of PaO_2) the execution of a matching task was improved in monkeys. The discrimination task used probably required controlled processing since the relationship between stimuli and responses varied during the task. We have found only one case in the literature where performance improved because of slight hypoxia. In an inter-group study, Kelman and Crow (1969) subjects exposed to a simulated altitude of 2,438 m performed a card sorting test faster.

Alterations in long-term memory, specifically episodic memory, have rarely been observed, and always as an acute effect, never as long-term effect. Garrido

et al. (1995) discuss a case of transitory global amnesia strongly related to the symptoms of AMS and High-Altitude Cerebral Edema during the ascent to a summit above 8,000 m. Litch and Bishop (1999) describe two cases with the same alteration, both at moderate altitude (4,400 m and 3,750 m). Their subjects were disoriented at spatial and personal level, and unable to remember autobiographic episodes. In both cases, the disorder occurred without severe AMS and ceased with relative quickness when subjects descended 450 m and 760 m, respectively.

Spatial Memory, Short-Term Memory and Attention

The effects of hypoxia on rats have also been evaluated with respect to localization and spatial memory. Using the MWM, Shukitt-Hale et al. (1994) found that acute exposure to a variable altitude between 5,500 m and 6,400 m produced an altitude-dependant dysfunction to spatial memory. The effects were observed in the speed, distance covered, and time taken to reach a hidden platform. It should be highlighted that it was learning, rather than retrieval, that was impaired. The task was learned under hypoxia and retrieved under normoxia. The effect was only noticeable beyond 5,500 m so this seems to be the lower threshold for dysfunction in rats. In general, results were less significant when evaluated after exposure than during exposure. Significance was also greater when performance was evaluated after 6 hours of exposure than after 2 hours. In contrast, there was a more significant impairment in the execution of the first trial as compared with the second, which suggests a greater dysfunction to the learning of new abilities than to the retrieval or execution as observed above. Gozal et al. (2001) have observed similar results with the MWM applying intermittent hypoxia. Nelson (1982) evaluated spatial memory in 20 mountaineers during a 35-day climb to Mount McKinley (6,193 m). At 3,800 m no impairment was noticed but at 5,000 m results were significantly inferior.

In a short-term memory task, Crow and Kelman (1971) had 86 subjects learn lists of numbers at different simulated altitudes. Indications of dysfunction ($p = .07$) were only found at the highest altitude (3,658 m). In a later study, these authors obtained identical results using a free recall task (Crow and Kelman, 1973). Bartholomew et al. (1999) had 72 aircraft pilots make radio calls at 606 m, 3,788 m, and 4,545 m. Half of them required a high memory capacity (4–5 bits). Significant deficit was only observed at 4,545 m in those trials that required a high capacity of memory. Virués et al. (2002) observed a slight but significant reduction in short-term memory established through Weschler's Digit Span associated with

climbs to 4,800 m and 4,600 m. Apparently the lower threshold of altitude to produce memory dysfunction is above 3,500 m, in agreement with observations in animals (Shukitt-Hale et al., 1994).

The effects of altitude on attention capacity have rarely been described. Evans and Witt (1966) found impairment in their subjects with the Digit Symbol test from the Wechsler Adult Intelligence Scale at 4,200 m. Berry et al. (1989) also report the finding of impairment in the Digit Symbol test after exposure to simulated altitudes between 3,000 m and 5,000 m. Two recent studies have produced convergent results. Stivalet et al. (2000) noted that the results in a task of visual search (detecting a target among a group of distractors) were reduced after 8 hours of moderate hypoxia. Specifically, the authors concluded that hypoxia delayed the serial-attentional processing while the parallel-preattentional remained unaffected. In a field research, Bonnon et al. (2000) monitored the attention performance of one expedition that remained 16 days between 2,000 m and 5,600 m and took 2 more days to climb to 6,440 m and another that stayed 21 days at 6,542 m. They employed a modified version of the Fort Test with two levels of complexity. Attention impairment was only observed in the second expedition. The more difficult the task was, the greater the differences with the control group.

Theoretical Issues

There has been disagreement over the permanence of cognitive impairment induced by exposure to extreme altitude. In this respect, the effects on memory and motor capacity have been discussed. Kramer et al. (1993) recorded a reduced learning rate compared with controls 2 weeks after an ascent to Mount Denali (6,194 m). Similar results were obtained in COPD patients (Grant et al. 1987). Cavaletti et al. (1987) made records 75 days after returning from Mount Satopanth (7,075 m) and noticed that a memory reduction of 19% right after the ascent was then 14% lower than the pretest levels. Unfortunately, these authors do not specify the memory test they used nor the significance of their results. Furthermore, 12 months after an ascent to Mount Everest (8,848 m) Townes et al. (1984) found that the difference in number of attempts to reach the criterion in the WMS was still significant ($p < .05$). This effect ceased when measures were obtained 24 months later. It is noticeable that this research was performed in an ascent without supplementary oxygen. Eight world-class mountain climbers were subjected to a multitude of neuropsychological tests by Regard et al. (1989). These authors used as memory tests, the Digit Span and Histories from the Wechsler Adult Intelligence

Scale, an audio-verbal test, a visual learning test, and the reproduction of the Rey Complex Figure. Their results showed that two subjects displayed two standard deviations (*SD*) in verbal memory below a control group of low-altitude climbers whereas the other three were one *SD* below the control group. With respect to spatial memory, five subjects were two *SD* below the control group and one subject was one *SD* below the control group. This research is often referred to in order to support the existence of permanent damages. Nevertheless, the comparison with the control group is difficult to assess. In addition, we have observed that the mean time elapsed since the last exposure to an altitude above 5,000 m was only 7 months for all subjects, and in any case never more than 11 months.

Clinical models of chronic hypoxia can help to adjust future research on the effects of altitude as well as to obtain indirect evidence from highly controlled studies. For example, correlations between 0.75 and 0.82 have been found between memory capacity and hypoxia in COPD patients (Huppert, 1982). More recently and in a highly controlled investigation, Stuss et al. (1997) have found correlations in these patients above 0.7 between PaCO_2 and every index in the WMS with the exception of immediate logical memory. The same applies to the correlations with the Benton Visual Retention Test or the free recall with either short or long retention interval, and the total recall from list A in the California Verbal Learning Test (CVLT). Not so with the Digit Span task or the Brown-Peterson task. Associations in the case of PaO_2 lead to similar conclusions with a lesser statistical significance. Likewise, in the case of OSA, similar dysfunctions to memory and learning of new information have been found with the WMS (Kales et al., 1985; Kelly et al., 1990; Lezak, 1995) and the CVLT (Salorio et al., 2002).

At this stage, laboratory research should establish the interaction between hypocapnia and hypoxia in the account of the effect of altitude on memory. For this purpose, Berry et al. (1989) induced a hypoxia equivalent to 5,000 m in several subjects maintaining PaCO_2 (isocapnia) constant. No memory dysfunctions were found using SRT, a test sensitive to long-term verbal memory, whereas investigations made without controlling isocapnia did find significant differences in memory tests conducted in laboratory (Crow and Kelman, 1969; Shephard, 1956). Memory dysfunctions can be the specific result of hypocapnia although one can object (a) that hypocapnia does not result in an increased brain hypoxia below 8,000 m or a PaO_2 equivalent to 30 mmHg (West, 1984); and (b) that the SRT is a long-term memory test not used in other research incapable of distinguishing hypoxia from

hypocapnia. Additionally, the study by Berry et al. (1989) used %SaO₂ as independent variable (IV) with four conditions: 100%, 90%, 85%, and 80%. If we consider that 80% SaO₂ is equivalent to an altitude of 4,500 m the conclusion is that only one field investigation has found effects on memory at this or a lower altitude (McFarland, 1971). In this respect, Phillips and Pace (1966) did not find differences in short-term memory in their subjects compared with pretest levels either during or after a short exposure to 3,800 m, Sharma et al. (1975) performed immediate memory measurements 1, 10, and 13 months after ascending to an altitude of 4,000 m without finding significant differences with the pretest at any of these moments. The same can be concluded from the cited studies by Crow and Kelman (Crow and Kelman, 1971, 1973).

Against the hypothesis that memory dysfunctions are only observed after stays above 4,500 m we can object that these designs frequently ignore the interaction between altitude and task difficulty, so that similar results are found at lower altitudes if tasks are harder (Bartholomew et al., 1999; Bonnon et al., 2000; Cahoon, 1972; Cudaback, 1984).

Measurements of a simulated ascent to 8,848 m for 40 days inside a hypobaric chamber (Hornbein et al., 1989) taken in laboratory conditions showed that the SRT detects significant damages when the altitude is above 4,500 m. In this investigation the neuropsychological impairment was compared between two groups, one exposed to extreme altitude inside a hypobaric chamber and another belonging to an expedition to Mount Everest (8,848 m). The results of the SRT were not convergent since damage was only found in the laboratory group. The results of the WMS indicated damages in long-term visual memory. Again we face the problem of IV. Although the barometric altitude in both cases was equivalent, no direct measurements of blood gases were made (PaCO₂, PaO₂, and %SaO₂) thus ignoring the fact that subjects with a low HVR suffer a greater hypoxia under an intense workload. This factor may be a source of confusion in the results because no comparison can be made between the physical activity of a group confined inside a chamber with another climbing to a peak of 8,000 m. It is unlikely that such results can be attributed to differences in stress or confinement factors (Farrace et al., 1999). With respect to differential hypoxia according to workload, Bakharev (1981) noted that his subjects displayed greater memory damages in the days of intense workload. This author did not take direct measurements of %SaO₂ although presumably the key factor for the greater alteration was an increased hypoxia due to intense workloads (Roach et al., 2000) especially considering that his subjects were work-

ers taken to 3,600 m. So they did not have such a high mean HVR as can be found in mountain climbers (Johnson and Rock, 1988).

Memory alterations due to high and extreme altitude have been replicated for decades. The effect becomes consistent in exposure to altitudes above 4,500 m although some conditions such as the workload or the level of HVR modulate hypoxemia. The effect is more solid in laboratory conditions. A group of investigations with actual ascents above 5,000 m found no conclusive results (Clark et al., 1983; Jason et al., 1989; Milne and Gray, 1983). Animal research has observed deficits in spatial memory due to hypoxia. This has been related to damages to the hippocampus resulting from hypoxia. This point has not been correctly replicated in humans. In most cases, researchers have administered immediate memory, and short and long-term verbal and visual memory tests. Isolated cases of global amnesia without posterior sequels have been observed. Impairment seems to affect specifically codification rather than retrieval. Motor factors do not seem to be involved in such deterioration. Several studies demonstrate that dysfunctions have an average duration after returning to lowland if exposure surpassed 7,000 m. The previous evidence is not enough to conclude that hypoxia due to extreme altitude triggers permanent memory impairment.

FUNCTIONS ASSOCIATED WITH FRONTAL LOBE: LANGUAGE, COGNITIVE FLEXIBILITY AND METAMEMORY

Several studies have identified frontal alterations after exposure to extreme altitude. Such alterations are related to the sensitivity of the frontal lobe and to slight hypoxia. Impairment in language and verbal fluency have been repeatedly reported. In an actual expedition that reached 6,800 m, Petiet (1988) verified alterations in abstract reasoning and verbal fluency by means of the Word Finding Ability Test both during and right after the ascent. Kennedy et al. (1989) registered significant reductions in grammatical reasoning and the Digit Symbol test in subjects after a simulated ascent to 8,400 m. Cavaletti et al. (1987) give details of alterations in verbal fluency after an ascent to the Himalayas that persisted 2 months after returning to lowland. Regard et al. (1989) used two fluency tests based on the production of words and figures corresponding to a category in a limited time. They found a considerable impairment in world-class climbers as compared with a control group after the former had spent an average of 7 months without climbing above 5,000 m. Using the Aphasia Screening Test, both West

(1984) and Hornbein et al. (1989) have observed alterations in language production and an increase in aphasic errors.

In the study by West (1984), records were collected a year later showing that impairment had reverted to baseline levels. Shipton (1943), and Garrido and Javierre (1996) have reported cases of transitory aphasia due to exposure to extreme altitude. Garrido and Javierre (1996) observed slight signals of high intensity in a subject's posterior lobe in an MRI carried out 40 days after the descent. Analyzing radio calls during an actual expedition to Mount Everest, Lieberman et al. (1995) found an altitude-dependant effect regarding articulatory alterations along with increased time to understand sentences. Similar results have been published in other cases of hypoxia not related to altitude. Fox et al. (1989) observed alterations in resonance, articulation, and phonation in 74% of patients with OSA and 53% of patients with COPD as compared with a 7% in normal cases. Discrimination analyses identified 96.3% of patients without apnea and 63% of patients with apnea. Kazora et al. (1999) found out a lower verbal fluency in patients with COPD by means of Letterfluency. Salorio et al. (2002) has obtained the same results in OSA patients with this test.

With respect to cognitive flexibility, Regard et al. (1989) noticed clear impairment in their subjects (world-class mountain climbers evaluated months after their latest ascent) through inference tasks of the Stroop Color and Word Test or the Wisconsin Card Sorting Test. These authors averaged the results in the Stroop Color and Word Test, the Wisconsin Card Sorting Test, and two other verbal fluency tasks in a variable called *cognitive flexibility*. This variable presented the most impaired results in the group studied compared with the other cognitive processes assessed. Two subjects showed a performance one *SD* below the controls while the remaining six subjects performed two *SD* below. Recently, Van Diest et al. (2000) have found similar results concerning the Stroop Color and Word Test under slight hypoxia in laboratory conditions. In this case, brain hypoxia was induced through hyperventilation, as a result of brain vasoconstriction dependent on hypocapnia. It should be noted that results were obtained at 90 %SaO₂, equivalent to 2,500 m. Such effect contradicts the apparently lesser vulnerability of the frontal lobe but in any case the presence of a strong hypocapnia predicts an intense central vasoconstriction that would increase the levels of increased tissue hypoxia in the brain above the degree of SaO₂ at peripheral level. Deficient performance in the Wisconsin Card Sorting Test has also been observed in patients with COPD (Crews et al., 2001), but not in patients with OSA (Salorio et al., 2002).

Metacognition or metamemory may also be impaired under hypoxia (Janowsky et al., 1989). An interesting study in this respect was conducted by Nelson et al. (1990). In an expedition to the Himalayas above 6,400 m, subjects took repeatedly the Facteretrieval2, a test which measures the retrieval capacity of short-term memory together with metacognitive aspects of the retrieved content (feeling of knowing what has been retrieved). Results showed that although no damages in the retrieval capacity of long-term memory were observed, a reduction took place in the *feeling of knowing* that remained a week after returning from the expedition. Clark et al. (1983) obtained similar results after a prolonged stay at 5,300 m. Subjects rated their own performance in several tasks as poorer than before the climb without an actual damage in the execution. Something similar has been noted in the case of insomnia associated with altitude. Great differences are observed between the hours subjects think they sleep, few or none, and the duration of actual sleep (Reite et al., 1975).

Higher processes such as language, cognitive flexibility, and metacognition may be impaired due to hypoxia. Regarding language, clear damages are noticed in verbal fluency and aphasic errors. In addition, several studies have reported reduced abstract reasoning, understanding and articulation. Difficulties in articulation and verbal fluency converge with impairments observed in chronic hypoxia disorders (OSA and COPD). Published studies have confirmed such deterioration above 6,400 m. Although no long-term language impairment occurred after returning to lowland. Reductions in cognitive flexibility and in resistance to interference have been replicated several times above 2,500 m. Alterations in various metacognitive processes have also been verified, and there is a general feeling of reduced confidence in performance in cognitive and memory tasks.

PERSONALITY, ANXIETY, AND HYPOXIA

Three aspects have been considered regarding the connections among personality, anxiety, and hypoxia: (a) which personality traits distinguish mountaineers from normal subjects; (b) which alterations in emotional states are associated with altitude; and (c) the influence of personality factors in adaptation to altitude.

Several investigations have highlighted that mountaineers display special personality traits. Magni et al. (1985) compared the results of the 16 personality factor questionnaire (16PF) of 22 mountain climbers from an expedition to the Himalayas with a control group. Significant differences were obtained in factors A

(reserved-outgoing), G (expedient-conscientious), and Q (relaxed-tense). Their results showed that mountaineers exhibit less anxiety, less social control, and greater emotional stability. Freixanet (1991) observed lower levels of neuroticism as measured by the Eysenck Personality Questionnaire when he compared subjects practicing risk sports with a control group. Other personality traits differing in mountaineers and subjects involved in risk activities is a higher total score in sensation seeking and a lower score in social conformity established by Zuckerman's Sensation Seeking Scale (Bushov et al., 1994; Freixanet, 1991; Jack and Ronan, 1998). Using the Zuckerman's Sensation Seeking Scale, Zarewski et al. (1998) noticed that the subscales of experience seeking and adventure seeking predicted involvement in high-risk sports and were significant in a posterior discriminating analysis between high-risk and low-risk sportsmen. Also, Noël-Jorand et al. (2001) argue that mountaineers are more stoical and introverted in comparison with other subjects.

Regarding the connection between emotional states and altitude, Ryn (1971, 1988) concluded, after observing 80 mountaineers, that at low altitude (below 2,500 m) emotional excitation or indifference were dominant while at high altitude (up to 5,500 m) euphoric-impulsive or depressive syndromes took place. Later, more systematic reports have tested that the level of anxiety established by the State-Trait Anxiety Inventory (STAI) increases with altitude and such increase correlates highly with impairment in the mood state (Bolmont and Abraini, 2001). Using the Profile of Mood States with a group of subjects that stayed at 3,630 m for a week, Shukitt-Hale et al. (1990) noticed how the scores in mood state became reduced. This author remarked that alterations occurred in an elevation-dependent fashion up to an altitude of 4,700 m (Shukitt-Hale et al., 1998) and that such impairment followed a different temporal pattern from that of the AMS (Shukitt-Hale and Banderet, 1988). However, a subsequent study Shukitt-Hale et al., (1991) demonstrated that after 7 hours at 4,700 m, 13 measures of the mood state correlated highly ($r = .77$) with the brain factor (AMS-C), measured by the Environmental Symptoms Questionnaire, a scale that quantifies the degree of AMS.

One important related issue is the influence of personality and mood state in adaptation to altitude. Early studies (Finesinger et al., 1947; McFarland and Barach, 1937; Waldfogel et al., 1950) showed that subjects diagnosed with psychoneurosis display a lower PAO_2 and more serious breathing problems in comparison with control subjects during adaptation to conditions of acute hypoxia. Missoum et al. (1992) studied a group of 100 climbers from an expedition to the Himalayas. After the expedition, subjects were divided between susceptible

and nonsusceptible to AMS. It was demonstrated that susceptible subjects had a higher level of anxiety-trait in the STAI before the expedition and a higher level of anxiety-state after the latest ascent. Bushov et al. (1994) exposed their subjects to a simulated altitude equivalent to 3,500 m to distinguish between normals and mountaineers. It was observed that the level of neuroticism, whose group levels were lower in mountaineers, moderated the physical adaptation to altitude. Reduced levels of neuroticism correlated with reduced levels in the symptoms of AMS. Nicolas et al. (2000) obtained negative correlations between fatigue, a symptom of AMS, and emotional stability as measured by 16PF (factor C), in the experiment Everest-Comex 97 (Operation Everest III), in which volunteers remained inside a hypobaric chamber for 31 days exposed to altitudes up to 8,848 m. In another study based on this project, Bolmont et al. (2001) found high and positive correlations between anxiety-trait established by STAI before the ascent, and CRT during hypoxia. Negative correlations were obtained regarding Factor A (reserved-outgoing), and Factor G (expedient-conscientious) from the 16PF with the CRT. No similar connections could be found with more complex cognitive or motor activities. These authors conclude that the influence of anxiety under hypoxia is only exerted on stimulus-response tasks but not on more complex cognitive or psychomotor tasks.

According to this information, it is advisable to include mountaineers in control groups since this difference between groups could produce mixed results. It is also desirable to make *a posteriori* controls of the influence of the neuroticism variable, emotional stability or anxiety-trait, depending on the test used. A second methodological consideration relates to the association between anxiety-trait and memory processes. Specifically, positive correlations have been established between anxiety and an increase in implicit memory (McLeod and McLaughlin, 1995), while negative correlation has been found between anxiety and working memory (Hopko et al., 1998; Ikeda et al., 1996). For this reason, this variable should be considered in investigations of relationships between altitude and memory, which is usually disregarded in follow-up analyses. Similar effects of anxiety have been found in RT and motor tasks (Bolmont et al., 2000).

With the caveat of the limited research available, we observe a remarkable impact of personality traits associated with anxiety (emotional stability, anxiety-trait, or neuroticism) and differences in HVR as individual differences moderating the effects of exposure to altitude. As noted above, emotional stability is associated with a greater physical adaptation to altitude as long as fatigue and AMS symptoms are concerned. This variable may

regulate a different pattern of effects although the results published by Bolmont et al. (2001) suggest that the impact of anxiety is limited to simple stimulus-response tasks.

METHODOLOGICAL ISSUES: THE INDEPENDENT VARIABLE

The effects of hypoxia have been considered in different scientific contexts. Since the beginning of the 20th century, the performance of human beings at extreme altitude has been taken into account in studies involving the aeronautical domain, hypobaric chambers and mountain ascents. Likewise, there are various clinical models of chronic hypoxia whose effects resemble acute hypoxia (COPD and OSA). Slightly different results have arisen from every model but they share a general pattern of impairment. Investigations on the effects of altitude in flights differ from those in mountains in the exposure to higher altitudes, much greater rates of ascent, and use of supplementary oxygen. Neither the physical fatigue nor the stress associated with mountain climbing is present in flights. In this respect, methodological concerns might be arising in comparing laboratory studies—*in vitro*- and real-ascent studies—*in vivo*-. *In vivo* studies sometimes confuse the altitude variable with other factors that systematically covariate with altitude (Table 3). However, as shown above, there is a general tendency to obtain converging results. With some exceptions that have already

been discussed (Hornbein et al., 1989) there are no investigations that make the two types of studies readily and reliably comparable.

The most frequent IV in this field is altitude in meters above sea level. The correct manipulation of this IV is an added difficulty in these studies. What we call *altitude* is a set of factors with unknown specific effects on ANI in particular. For example, low environmental temperatures covariate with altitude exposition in real-ascent studies. Low temperature induces peripheral vasodilatation and thyroid gland hyperfunctioning, but the impact of these factors has not been measured. Similarly, inadequate nourishment, anorexia, and weight loss are associated to prolonged hypoxia in high altitude (Hackett and Rennie, 1976; Milledge, 2002; Timiras et al., 1957). They are error variables of unknown influence on ANI. A revision is made of such factors in Table 3. Possible ways of controlling this sort of variability are also pointed out.

Geographic altitude is a variable that should be divided into an array of factors in order to assess its influence on the human organism. Its main effects are hypoxia, hypoxemia, and hypocapnia. Other environmental and individual variables that appear inadvertently in the IV—imprecisely called *altitude*—may have an influence too. Ignoring such factors impinges seriously on the finding of differences given the reduced number of subjects in these investigations and the minimal degree of impairment (Clark et al., 1983; Jason et al., 1989). Whenever

Table 3. Possible Confounding Variables in ANI Research

Confounding variable	Studies affected		Possible way of control
	Setting	Design	
Environmental variables			
Variability ^a in the reduction of P_B	IV	IG, WS	Consider P_B as independent variable
Reduction of gravity	IV	IG, WS	—
Environmental temperature	IV	IG	Simulate temperature conditions in a nonascent CC
Over-stimulation	IV	IG	Simulate stimulation conditions in a nonascent CC
Alterations in the circadian rhythm	IV	IG	Simulate bio-rhythmic conditions in a nonascent CC
Confinement in simulated ascents	IVT	IG	Consider a nonconfinement control group
Physiological variables			
Variability ^a in the reduction of PaO_2	IV, IVT	IG	Consider % SaO_2 as independent variable in IVT research
Variability ^a in the reduction of % SaO_2	IV, IVT	IG	Consider % SaO_2 as independent variable in IVT research
Variability ^a in the reduction of PaCO_2	IV, IVT	IG	Constancy control or co-variance analyses in IVT research
Variability ^a in HVR	IV, IVT	IG	Constancy control or co-variance analyses
Variability in exercise	IV	IG	Simulate exercise conditions in a nonreal-ascent CC
Weight reduction related to altitude	IV	IG, WS	Co-variance analyses in WS research
Psychological variables			
Personality traits	IV, IVT	IG	Match relevant personality traits in control subjects

Notes. CC = control group; IV = *in vivo* (i.e., real ascent studies); IVT = *in vitro* (i.e., hypobaric chamber studies); IG = inter-groups; WS = within-subjects.

^aVariability related to the same altitude.

these measures have not been controlled, associated impairment has been found at lower altitudes (Fowler et al., 1987).

Other Methodological Shortcomings

The use of within-subject designs makes it impossible to determine potential effects of practice and the passing of time. This could be overcome by including control groups whose dependent measures were registered at the same time that actual expeditions, or in the case of moderate altitude, repeating climbs at low altitude to check whether damages occur in successive ascents. Virués et al. (2002) exposed subjects to climbs to altitudes around 4,500 m and repeated the experiment after a week. Results showed that impairment reverted during the rest period and reappeared in the second ascent, discarding the effect of practice or the passing of time.

Other difficulties are derived from the setting in which tests are recorded. Evaluations made days before climbing a peak are likely to be biased by the high levels of anxiety and reduced attention. Few studies have attempted to dissociate the effects of attention from others (Van Diest et al., 2000; Virués et al., 2002). In addition, in High Mountain, tests are often done by nonspecialists so that the reliability of results can be reduced. Experimenters can also affect negatively the results, especially through emotional variables. Barach (1944) exposed a group of students to an atmosphere of 13 %O₂ (3,660 m) for 3 hours. The subjects were asked about their euphoric and depressive tendencies during the experience. Such tendencies appeared with a greater frequency when the experimenter was an attractive woman than when a man carried out the test. Other psychosocial factors such as result expectations, social deprivation or leadership have not been considered in research on the effects of altitude. Several methodological limitations in this respect are summarized in Table 4.

FUTURE DEMANDS OF EVALUATION OF ALTITUDE NEUROPSYCHOLOGICAL IMPAIRMENT AND A FEW USEFUL GUIDELINES

The present study has revised 70 years of investigations on the balance between mind and brain during exposure to altitude in a broad sense. Throughout this article, methodological shortcomings and difficulties of comparison have been observed in many studies. However, there is an unquestionable impairment of motor, perceptive, memory, and executive functions as an acute effect and at least with middle-term duration. Future research will benefit from more accurate tests since the effects of ANI are slight. Improvement in control groups and a better definition of the IV are further issues to consider.

Another aspect that has been overlooked is the possible set of psychological strategies to reduce hypoxia, and eventually ANI. Savourey et al. (1995) reported that AMS symptoms were associated with breathing pattern more often than with HVR. Conversely, Saul et al. (2002) demonstrated that voluntary hyperventilation can relapse hypoxia in subjects that do not present this response spontaneously. In this respect we claim that biofeedback procedures can help in learning protective breathing patterns at sea level. As a result, useful training programs could be developed to lessen the damaging effects of hypoxia. This idea has been put into effect for other respiratory diseases. Fried (1995) has successfully used biofeedback from a capnometer, an oxymeter and occasionally electromiographic biofeedback from the thorax in order to control respiratory alkalosis in patients with chronic hyperventilation. The interest of the data supplied by the capnometer is obvious since it allows to produce a voluntary ventilatory response before reaching the damaging effects of hypocapnia. Hillsman (1996a) developed a protocol of visual biofeedback to teach breathing patterns. According to Hillsman's procedure, the tidal volume is shown on axis Y and the respiratory rate on axis X on a screen. The patient's real time breathing is displayed and at the same

Table 4. Frequent Methodological Difficulties in Research on ANI

Methodological issues
Effect of practice and passing of time in within-subject investigations.
Effect of anxiety in pretest registers.
Posttest measurements too close to descent to evaluate lasting effects or too distant to evaluate acute effects.
Control group does not perform a similar amount of physical exercise.
Control group is made up of nonclimbers.
Ceiling effect due to inadequate tests, insufficient maximum altitude, and inadequate stay.
Most investigations referred to adult western males with a high socioeconomic level and high physical fitness.
No dissociation between the effects of hypoxia and hypocapnia.

Table 5. Suggested Battery to Evaluate ANI

Assessment goal	Assessment instruments
ANI	Aphasia screening test Burchle's selective reminding test Computerized measures of reaction time Digit span test Finger tapping test Perdue pegboard Test Stroop color-word test Symbol digit modalities test Trail making test Verbal fluency test Visual cancellation tasks Wechsler memory scale
Acute mountain sickness symptoms	Environmental symptoms checklist Lake Louise AMS scoring system

Note. The instruments were gathered in regards to four criteria: (a) known psychometric properties, (b) being affected by altitude in at least to studies, (c) low practice and ceiling effects, and (d) suitability of administration by nonexperts. Partially adapted from Echemendia, R. J., and Julian, L. J. (2001). Mild traumatic brain injury in sports: Neuropsychology's contribution to a developing field. *Neuropsychol. Rev.* **11**: 69–88.

time the prescribed breathing pattern is shown. Through successive approximations the patient finally adopts the ideal pattern. Similar procedures have been employed in COPD (Hillsman, 1996b), asthma (Dahmea, 1996), and other respiratory diseases. The efficiency of these methods in altitude-related hypoxia will have consequences in the prevention of AMS and ANI, at least at moderate and high altitude.

Up to now, the neuropsychological tests applied to investigations on ANI have been diverse. Some tests with inadequately established psychometric properties have been applied. This circumstance limits an appropriate description of the sample and the variability of the measure in within-subject investigations, which are most frequent in the field. We partially subscribe to the core battery proposal by Echemendia and Julian (2001) for the evaluation of sports-related mild traumatic brain injury. The tests included by these authors in their battery have been widely applied to neuropsychological damages related to sports and have well-known psychometric parameters. These tests also meet the practical requirements that increase the effectiveness in the sports environment: little ceiling and practice effects, could be administered by nonexperts and could be administered in a short time (Echemendia and Julian, 2001). However, given that the mild traumatic brain injury can be due to very different brain dysfunctions, we strongly advocate specific tests for research on altitude. These tests meet the conditions of hav-

ing been affected by altitude on at least two occasions, of having known psychometric properties, and meeting the above-mentioned practical requirements of Echemendia and Julian (2001). Table 5 shows a proposal for such a battery.

The construction and generalization of a standardized protocol will facilitate an ever difficult comparison among investigations. Moreover, the addition of neuropsychological criteria to those of structural and functional brain damage may allow advanced assessment of patterns of functional impairment due to altitude. Neuropsychological assessment at high and extreme altitude may extend the knowledge on the theoretical relations among hypoxia, brain, behavior and cognition.

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